

Backlog Buddy

Demonstration question paper generated from available previous papers - for practice only

B.Tech. I Semester Examination, 2025

Regulation: R20 | University: JNTUK

Digital Logic & Computer Organization (21CS34)

Time: 3 hours Max. Marks: 60

Answer one question from each unit. All questions carry equal marks.

UNIT - 1 [Basic Structure of Computers]

- a) Explain register transfer language and micro operations with examples.
- b) Explain the basic structure of a computer with a neat diagram.

UNIT - 2 [Microprogrammed Control Unit]

- a) Compare hardwired and microprogrammed control units.
- b) Explain microinstruction formats and control memory addressing.

UNIT - 3 [Computer Arithmetic]

- a) Explain Booth multiplication algorithm with an example.
- b) Explain floating point addition and subtraction with an example.

UNIT - 4 [Input / Output Organization]

- a) Explain interrupt driven I/O and DMA data transfer.
- b) Explain the different modes of DMA transfer.

UNIT - 5 [Memory Organization & Pipelining]

- a) Explain direct mapped, associative and set associative cache mapping.
- b) Explain instruction pipelining and the various pipeline hazards.

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